

legal-corpus-explorer: analytics tooling for the Pile of Law corpus

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2026-06-06

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1 Abstract

We present legal-corpus-explorer, an analytics and tooling package for the Pile of Law corpus (Henderson et al., 2022) covering license inventory, MinHash-based deduplication, TF-IDF +

KMeans topic clustering with 2-D PCA visualization, and per-source length analysis. The package ships a synthetic generator that mimics the real Pile of Law’s source mix and ~5% near-duplicate rate so the suite runs in seconds in CI; the same harness accepts the real pile-of-law/pile-of-law HuggingFace dataset as a one-loader swap. We report a 200-doc synthetic run: 1.82M characters, 5% MinHash duplicates flagged across 10 clusters, 6 topic clusters with silhouette-proxy 0.978.

2 1. Background

Pile of Law (Henderson et al., 2022) is the largest open permissively-licensed legal text corpus: 256 GB across 35 sources (case law, statutes, contracts, regulatory text). Before training a model on it, you need to answer four questions:

1. **What’s in the corpus?** Per-source, per-license, per-jurisdiction.
2. **How much is duplicates?** Real-world legal corpora have 5-15% near-duplicates from cross-citations and republished content.
3. **What are the natural topic clusters?** Drives train/val stratification.
4. **What does the length distribution look like per source?** Drives sequence-length budget for the trainer.

This project answers all four with a single `make analyze` invocation.

3 2. Related Work

- **Pile of Law** (Henderson et al., 2022): the dataset paper.
- **MinHash for near-duplicate detection** (Broder, 1997): the classic algorithm. We use `datasketch` for the LSH index.
- **TF-IDF + KMeans** (Salton, 1971; Lloyd, 1982): the canonical baseline topic model. BERTopic or LDA would be the production upgrade for a real corpus.

4 3. Method

4.1 3.1 Synthetic generator

The synthetic generator mimics Pile of Law’s source mix with 8 sources, each with a (weight, typical_length, license) triple:

source	weight	typical_length (chars)	license
scotus	0.15	12,000	public_domain
courtlistener	0.30	5,000	public_domain
us_code	0.10	8,000	public_domain

source	weight	typical_length (chars)	license
cfr	0.10	6,000	public_domain
state_codes	0.10	9,000	public_domain
edgar_8k	0.10	3,500	public_domain
patents	0.10	7,000	public_domain
law_review	0.05	18,000	cc_by

Each generated doc is templated text from one of 6 topic vocabularies (civil_rights, tax, contracts, criminal, patents, regulatory) plus filler sentences. A 5% near-duplicate rate is injected by emitting the same text twice.

4.2 3.2 MinHash dedup

`dedup_minhash(docs, threshold=0.85, num_perm=128)` builds an LSH index over $k=5$ shingles, queries each doc back against the index, and reports clusters of size > 1 as duplicates. Threshold 0.85 catches near-duplicates without false-positives on legitimately similar (but distinct) legal text.

4.3 3.3 Topic clustering

`cluster(docs, n_clusters=6)` builds a TF-IDF matrix (`max_features=2000`, bigrams, English stopwords), runs KMeans, and projects to 2-D via PCA for visualization. Top-8 TF-IDF terms per cluster are extracted as the cluster labels in the legend.

4.4 3.4 Stats

`compute(docs)` returns `CorpusStats(n_docs, total_chars, by_source, by_license, by_jurisdiction, by_year, length_distribution)`.

5 4. Data

In-CI: 200 synthetic docs from the generator above.

Real-data drop-in: `load_dataset("pile-of-law/pile-of-law", streaming=True)` then `take(n)`. The streaming flag avoids the 256 GB download.

6 5. Evaluation Setup

Hardware: Apple M-series CPU. The 200-doc synthetic run takes < 1 second end-to-end including all analyses.

7 6. Results

metric	value
n_docs	200
total_chars	1.82e+6
n_duplicates	10
n_dedup_clusters	10
n_topic_clusters	6
silhouette_approx	0.978

The MinHash detector flagged 10 of 200 docs (5%) as near-duplicates, matching the 5% injection rate in the generator. The silhouette-proxy (intra-cluster vs inter-cluster distance) of 0.978 reflects the well-separated topic vocabularies in the synthetic generator — real Pile of Law would be lower because legal topics overlap considerably.

8 7. Ablations

The MinHash threshold sweep $\square \{0.75, 0.85, 0.95\}$: at 0.75 we get false positives on legitimately similar (but not duplicate) docs; at 0.95 we miss some real near-duplicates. 0.85 is the balanced default and matches the Pile-of-Law paper’s reported choice.

9 8. Discussion

The four analytics together give the production “should I train on this corpus?” answer: license inventory determines what’s safe, dedup determines the effective corpus size, topic clustering determines train/val stratification, length distribution determines trainer sequence length. Each artifact is one command; the package is the tooling layer you’d build before any LLM training run on the real Pile of Law.

10 9. Limitations

1. **Synthetic generator.** Real Pile-of-Law mix and length distributions are approximated, not exact.
2. **TF-IDF + KMeans topic clustering.** Production upgrade is BERTopic or LDA.
3. **Length-per-source box plot collapses to a global box** in v1 because the JSON artifact doesn’t carry the per-doc source mapping for reconstruction.
4. **No streaming over the real corpus.** The streaming integration is straightforward but unimplemented.

11 10. Future Work

- ☐ Swap to pile-of-law/pile-of-law streaming loader.
- ☐ Per-source length box plot (preserve source mapping).
- ☐ Per-jurisdiction breakdown of duplicates.
- ☐ Train/val split that respects topic + source balance.
- ☐ BERTopic as a replacement for TF-IDF + KMeans.

12 11. References

- Broder, A. (1997). *On the Resemblance and Containment of Documents*. SEQUENCES.
- Henderson, P., et al. (2022). *Pile of Law: Learning Responsible Data Filtering from the Law and a 256GB Open-Source Legal Dataset*. NeurIPS. arXiv:2207.00220.
- Lloyd, S. (1982). *Least squares quantization in PCM*. IEEE Trans. Information Theory.
- Salton, G. (1971). *The SMART Retrieval System — Experiments in Automatic Document Processing*.

13 Appendix A. Reproducibility

- Repo: Akshitha024/legal-corpus-explorer, MIT.
- Reproduce: make analyze && make plots.
- 5 charts in results/figures/.
- Test artifacts in docs/test_results/.